Let $\Omega = [0,1]^d$ be a prob space
d is the number of sample sites.

$\sigma \sim \mathcal{N}^+(0,1)$
$\mu \sim \mathcal{N}(0,1)$
$\mathcal{N}(x \mid \mu, \sigma)$
] sample site $d = 2$.

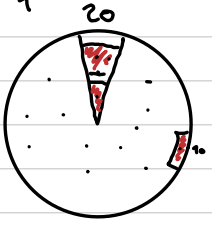
Need to introduce weighted traces.

```
x := 0
c := 1
while true
    c := // 3
    x := x + 2 · flip() × d
```
] d is countably infinite.

Idea is same as operational semantic 1.3
have rng list $[0,1]^\omega$

A program of type X determines
result : $\Omega \to X$ and weight : $\Omega \to \mathbb{R}^+$
↳ probably CDF

A frequentist semantic of a bayesian model:



we want prob of hitting shade region U.

k throws give mc integration: $\to P_n = \frac{\sum 1_{res(\omega_i) \in U} \cdot weigh(\omega_i)}{\sum_{i=1}^{k} weight}$
(why do we need the weights)

Law of large number (expectation of an indicator function.)

$$P_n \to P \quad as \quad n \to \infty \qquad (2.6)$$

$Prog_1 = Prog_2$ if $P_n^1 \to P \wedge P_n^2 \to P$.

2 problem
1. why integrals exist
2 we want to compose so we can do $(Prog_1 \circ Prog_2)(x)$.

σ-algebra for $\sigma(x)$ it holds that
$\emptyset \in \sigma(x)$
$a \in \sigma(x) \Rightarrow a^c \in \sigma(x)$
$B := (a_i \in \sigma(x))_{i \in I \subset \mathbb{N}} \Rightarrow \bigcup_i a_i \in \sigma(x)$.

measure space $(X, \sigma(x), \mu)$ s.t. $\mu : \Sigma \to \mathbb{R}^+ \cup \{\infty\}$
$\wedge \; \mu(\bigcup_i a_i \in \sigma(x)) = \sum_i \mu(a_i)$
$\wedge \; \mu(X) = 1 \; ; \; \mu(\emptyset) = 0$

density $f : X \to [0,1]$ is measurable and μ is a measure on X. Then
$$\nu(U) = \int_\mu [x \in U] f(x) \, dx$$
is a measure with density f wrt μ (usually lesbegue measure).

Ex $(r, x) \mapsto r \cdot \exp(-x)$ is a conditional density (missing r) wrt lesbegue measure.

Talk structure

Take example 2.2.1 and translate to prob prog.

3 semantics

1 mathematical using density and measures
2 MC with rejection
3 MC with weights.

Go through computation with each

Normalizing corner case: 0 bad, ∞ sometimes ok.

The meaning of a program is its measure (can be infinite)

Not all infinite work ← Need exchangability introduced.
example is counting measure

Sample ≙ like prior
Observe ≙ take data entry unit
normalize ≙ turn into normal measure

```
normalize (
  let x = sample (bern (2/7)) in          # true
      r = if x then 3 else 10 :in
      observe 4 from dr poisson (r);      # NN ≙
      return (x)                          1/d! x^d exp(-x)
)
```

method $p(x|obs) = \frac{p(obs|x)\,p(x)}{p(obs)}$

X		
true	2/2 · 0.168	0.22
false	0.048	0.78

reject($\frac{\#[d \sim r]}{\#\text{total}} \to p(x = true \mid obs : 4$

importance sample (x = true, weight = 0.168)
(x = false, weight = 0.048)

$$\frac{\#[true] \cdot weight}{\#false \cdot 0.048 + \#true \cdot 0.168} \to$$

(0.22 : 0 : 77)

two case division by 0 ←
Bad model
normalizing constant ∞ ←
Unbound likelihood;

$1 : t \mapsto r \exp(-rt) = \exp(t)$

```
normalize (
1   let t = exp(1) :in
2       obs 1 from normal(1, (2π e^{2t})^{-1/2} / τ_t );
        return (t)
)
```

2: $\sqrt{\frac{\tau}{2\pi}}\, e^{-\frac{1}{2}(t-1)^2 \tau} = \sqrt{\frac{\tau}{2\pi}} = e^t$

$$\int_0^\infty e^t e^{-t}\, dt = \int_0^\infty 1\, dt$$

$$p(obs) = \int_0^\infty p(obs, t)\, dt$$

$$= \int_0^\infty p(obs \mid t)\, p(t)\, dt$$

$$= \int_0^\infty e^t e^{-t}\, dt = \infty$$

$$\int_0^\infty 1\, dt$$

Improp
let x = Lebesgue in score(f(x)); return (x)

f is a probability density $[0,\infty) \to [0,\infty)$

$\int_M [x \in U] \cdot f(x)\, dx = \nu(x)$ is a measur

Subexpression can unnormalized
program must be normalized

A program of type X
results : $\Sigma \to X$ weight : $\Sigma \to [0,\infty)$

prob space $\Sigma = [0,1]^\infty$

sample(bern (2/7))

Prog 1 let x = sample (uniform) in return (x < 2/7)
Prog 2 let x = sample (uniform) in return (x > 5/7)

x = 1/7 res =
Prog 1 (w = 1, true)
Prog 2 (w = 1, false)

$$U = \frac{\sum_{i=1}^{k} [result(w_i) \in U] \cdot weigh(w_i)}{\sum_{i=1}^{k} weight(w_i)}$$

$k \to \infty$

$$\frac{\int_\Sigma [result(w) \in U] \cdot weight(w)\, dw}{\int_\Sigma weight(w)\, dw}$$

C[[p]] := let x : flip() in if x then p else return(0)

$Prog_A$ return(42)	observe ≙ density
$Prog_B$ score(2), return(42)	score(r) : weigth ↑

C [[$Prog_A$]] = 0 : 0.5
42 : 0.5

C [[$Prog_B$]] = 0 : 1/3
42 : 2/3

Improper distributions → posterior (normalize input) ✗
→ prior (program subexpression) ✓*

New const score(r) weights by r.
observe x p is let r := p(x) in score(r)

Improper posterior

⋮

Lebesgue measure → ∞ evidence ✗ (every event equiprobable)

Improper prior

Def If $f : X \to [0, \infty]$ is measurable, μ is a measure on X, then

$$\nu(U) = \int_\mu [x \in U] f(x)\, dx$$

is a measure.

f is the density of ν wrt μ.

if ν is a prob. measure then $\nu(X) = 1 \Rightarrow$

$$\int_\mu [x \in X] f(x)\, dx = 1 = \int_\mu f(x)\, dx$$